

DS4420 Final Project Bayesian Model

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Objective

We created a Bayesian predictive model that predicts the likelihood of pet adoption based on the animal's identified key traits.

Setup

Read Data

```
pets <- read.csv("pet_adoption_data.csv")
head(pets)
```

```
##   PetID PetType      Breed AgeMonths  Color  Size  WeightKg Vaccinated
## 1   500   Bird    Parakeet     131 Orange  Large  5.039768         1
## 2   501  Rabbit    Rabbit      73  White  Large 16.086727         0
## 3   502   Dog Golden Retriever   136 Orange Medium 2.076286         0
## 4   503   Bird    Parakeet      97  White  Small 3.339423         0
## 5   504  Rabbit    Rabbit     123  Gray  Large 20.498100         0
## 6   505   Dog    Labrador      70  Brown  Large 20.986261         0
##   HealthCondition TimeInShelterDays AdoptionFee PreviousOwner
## 1                0                27          140            0
## 2                0                 8          235            0
## 3                0                85          385            0
## 4                0                61          217            1
## 5                0                28           14            1
## 6                0                87          301            1
##   AdoptionLikelihood
## 1                    0
## 2                    0
## 3                    0
## 4                    0
## 5                    0
## 6                    0
```

Quick Data Checks

```
table(pets$AdoptionLikelihood, useNA = "ifany")

##
##      0      1
## 1348   659

model_cols <- c(
  "AdoptionLikelihood", "AgeMonths", "Breed", "PetType", "HealthCondition",
  "Vaccinated", "AdoptionFee", "TimeInShelterDays", "PreviousOwner", "WeightKg"
)
missing_cols <- setdiff(model_cols, names(pets))
if (length(missing_cols) > 0) stop("Missing columns: ", paste(missing_cols, collapse = ", "))
colSums(is.na(pets[model_cols]))

## AdoptionLikelihood      AgeMonths      Breed      PetType
##              0              0              0              0
##   HealthCondition   Vaccinated   AdoptionFee   TimeInShelterDays
##              0              0              0              0
##   PreviousOwner      WeightKg
##              0              0
```

Preprocess

```
pets_scaled <- pets

# numeric features get scaled
pets_scaled$AgeMonths <- as.numeric(scale(pets$AgeMonths))
pets_scaled$AdoptionFee <- as.numeric(scale(pets$AdoptionFee))
pets_scaled$WeightKg <- as.numeric(scale(pets$WeightKg))
pets_scaled$TimeInShelterDays <- as.numeric(scale(pets$TimeInShelterDays))

# categorical features
pets_scaled$Breed <- as.factor(pets$Breed)
pets_scaled$PetType <- as.factor(pets$PetType)
pets_scaled$HealthCondition <- as.factor(pets$HealthCondition)
pets_scaled$Vaccinated <- as.factor(pets$Vaccinated)
pets_scaled$PreviousOwner <- as.factor(pets$PreviousOwner)

# sanity check for Bernoulli outcome
stopifnot(all(pets_scaled$AdoptionLikelihood %in% c(0, 1)))
```

Fit Bayes Model

```
library(brms)

## Warning: package 'brms' was built under R version 4.3.3
```

```
## Warning: package 'Rcpp' was built under R version 4.3.3
```

```
n_chains <- 4
n_iter <- 4000
n_warmup <- 2000

model <- brm(
  AdoptionLikelihood ~ AgeMonths + Breed + PetType + HealthCondition + Vaccinated +
    AdoptionFee + TimeInShelterDays + PreviousOwner + WeightKg,
  family = bernoulli("logit"),
  data = pets_scaled,
  chains = n_chains,
  cores = 4,
  seed = 42,
  warmup = n_warmup,
  iter = n_iter,
  refresh = 100,
  control = list(adapt_delta = 0.95, max_treedepth = 12)
)
```

```
## Running "C:/PROGRA~1/R/R-43~1.1/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 12.3.0'
## gcc -I"C:/PROGRA~1/R/R-43~1.1/include" -DNDEBUG -I"C:/Users/jjams/AppData/Local/R/win-library/4.3/
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Core:19
##          from C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/Dense:1
##          from C:/Users/jjams/AppData/Local/R/win-library/4.3/StanHeaders/include/stan/math/p
##          from <command-line>:
## C:/Users/jjams/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10
## 679 | #include <cmath>
##      |          ^~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-43~1.1/etc/x64/Makeconf:265: foo.o] Error 1
```

```
## Warning: There were 7996 transitions after warmup that exceeded the maximum treedepth. Increase max_
## https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded
```

```
## Warning: Examine the pairs() plot to diagnose sampling problems
```

```
## Warning: The largest R-hat is 3.08, indicating chains have not mixed.
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#r-hat
```

```
## Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#bulk-ess
```

```
## Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quant
## Running the chains for more iterations may help. See
## https://mc-stan.org/misc/warnings.html#tail-ess
```

```
saveRDS(model, "bayes_model_fit.rds")
```

Outcomes

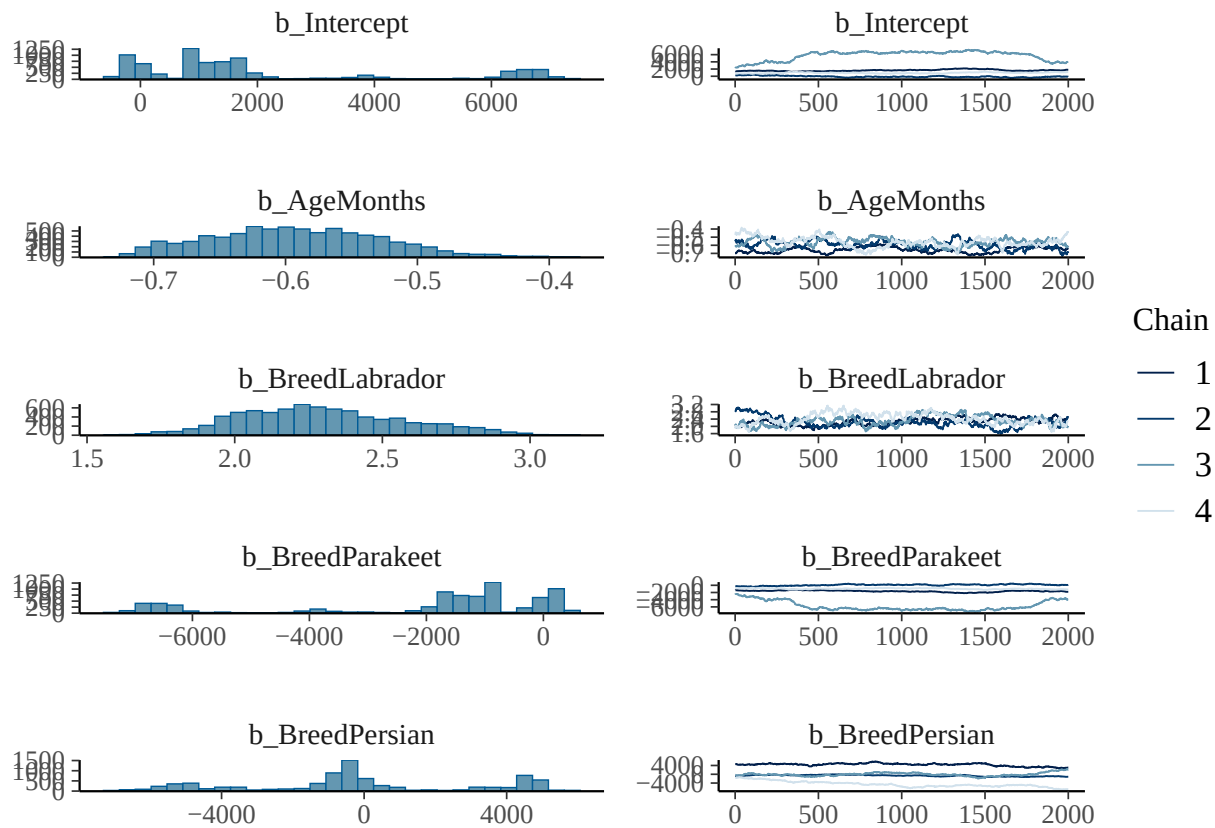
```
summary(model)
```

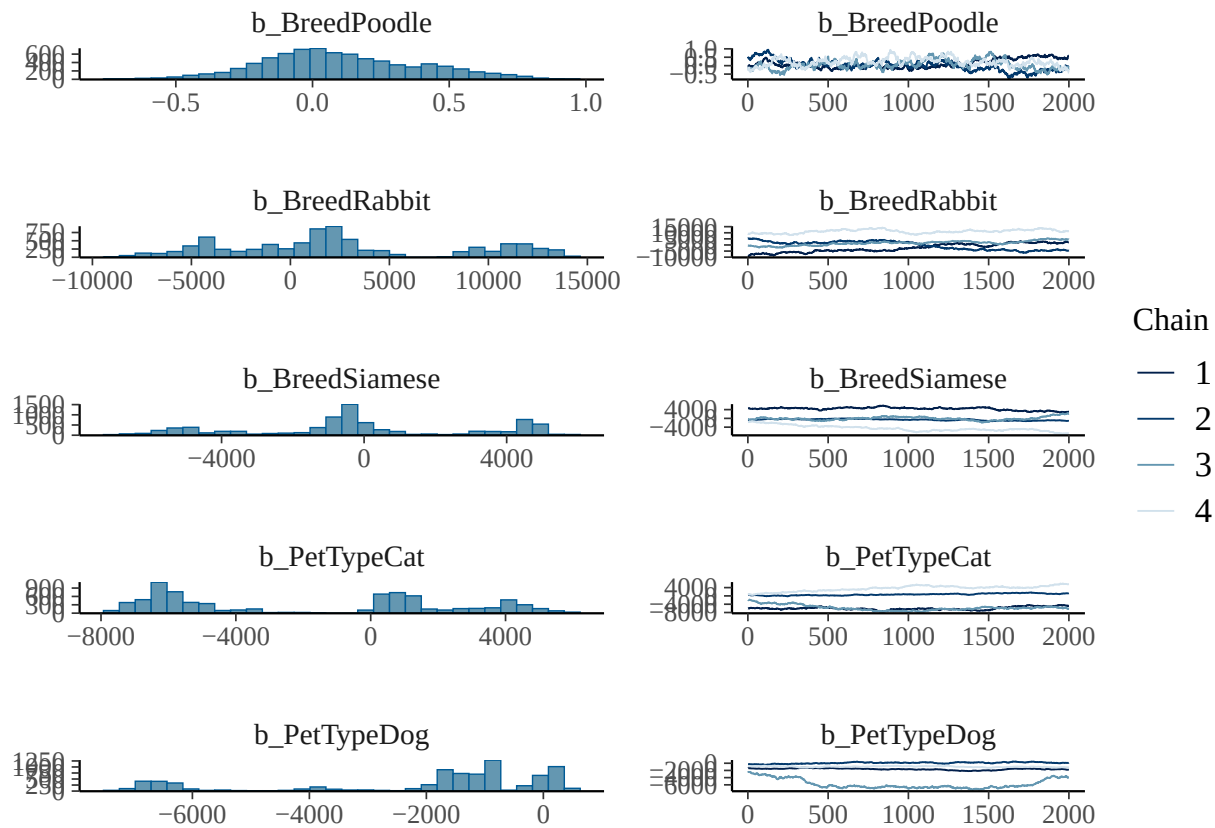
```
## Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
## careful when analysing the results! We recommend running more iterations and/or
## setting stronger priors.
```

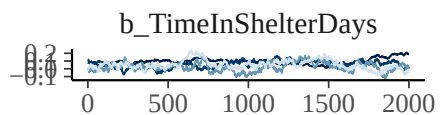
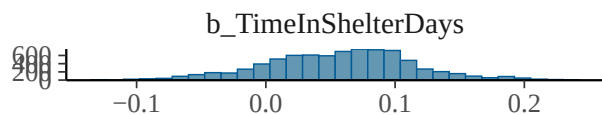
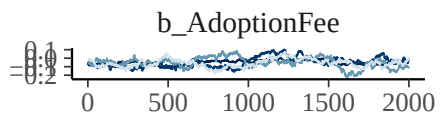
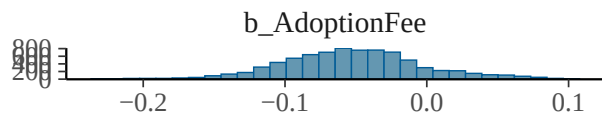
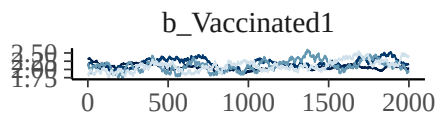
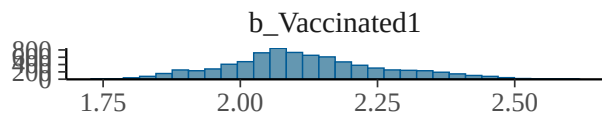
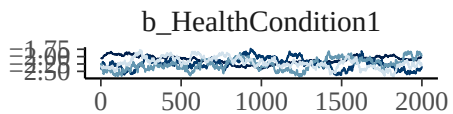
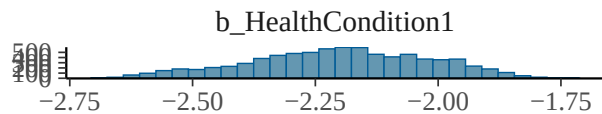
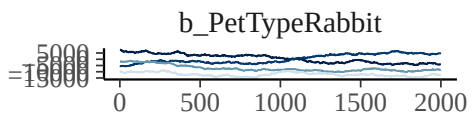
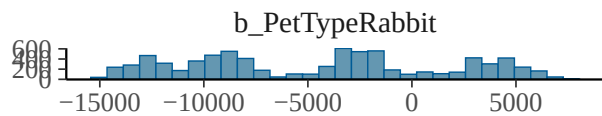
```
## Family: bernoulli
## Links: mu = logit
## Formula: AdoptionLikelihood ~ AgeMonths + Breed + PetType + HealthCondition + Vaccinated + AdoptionFee
## Data: pets_scaled (Number of observations: 2007)
## Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      2069.88   2332.11   -322.12  6909.92 3.08      5      22
## AgeMonths       -0.59     0.06    -0.70   -0.46 1.36      9      57
## BreedLabrador    2.30     0.28     1.82    2.88 1.17     18     85
## BreedParakeet   -2072.18   2332.09  -6912.25  319.76 3.08      5     22
## BreedPersian    -211.46   3259.62  -5934.03  4999.85 3.01      5     13
## BreedPoodle      0.11     0.29    -0.43    0.70 1.19     15     29
## BreedRabbit     2504.59   5829.26  -6871.73 13157.91 2.65      5     15
## BreedSiamese    -211.23   3259.63  -5933.77  5000.03 3.01      5     13
## PetTypeCat     -1860.87   4248.63  -7302.14  5088.63 2.26      5     14
## PetTypeDog     -2072.07   2332.10  -6912.14  320.32 3.08      5     22
## PetTypeRabbit   -4576.94   6121.95 -14125.13  5796.17 2.69      5     24
## HealthCondition1 -2.20     0.18    -2.57   -1.88 1.13     25    138
## Vaccinated1      2.12     0.14     1.86    2.43 1.15     21     95
## AdoptionFee     -0.05     0.05    -0.15    0.06 1.20     16     40
## TimeInShelterDays 0.06     0.06    -0.06    0.18 1.17     19     46
## PreviousOwner1  -0.12     0.11    -0.32    0.10 1.26     11     53
## WeightKg         0.01     0.04    -0.07    0.10 1.05     59    115
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Diagnostics (training quality)

```
plot(model)
```

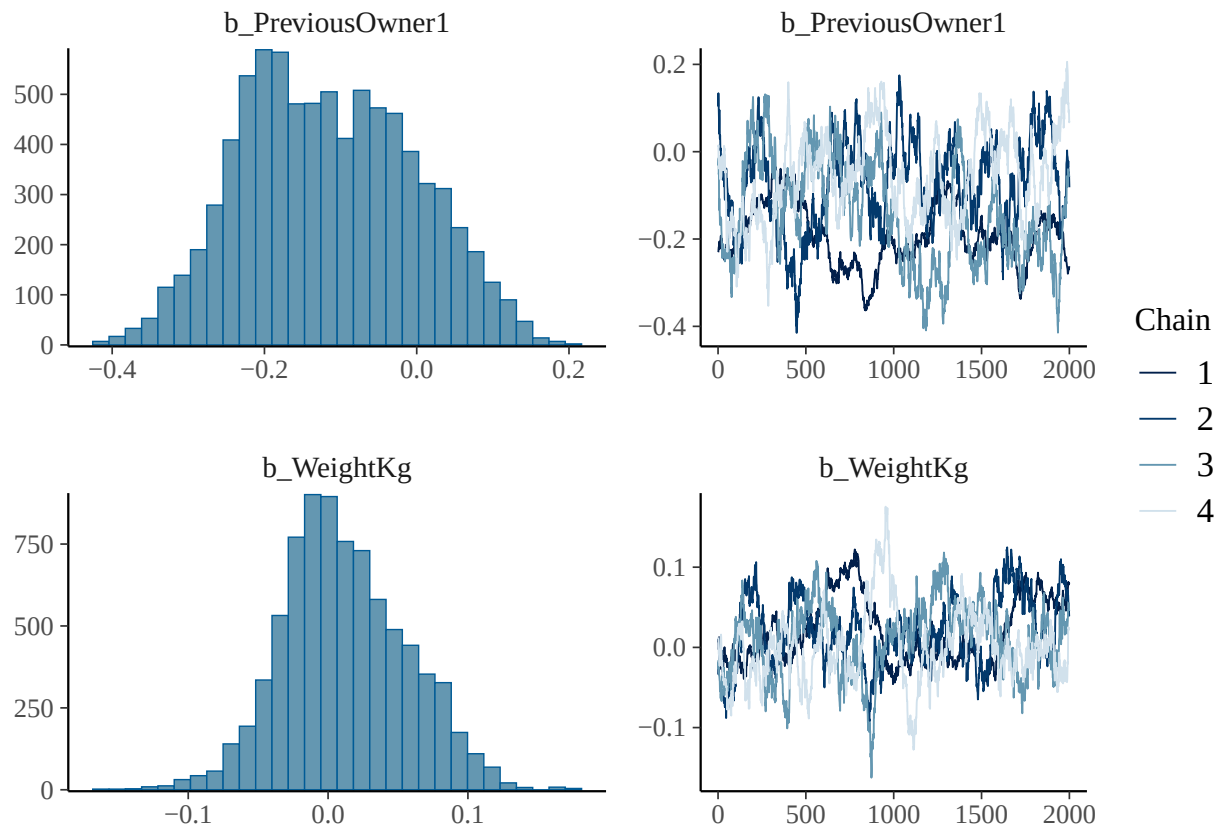




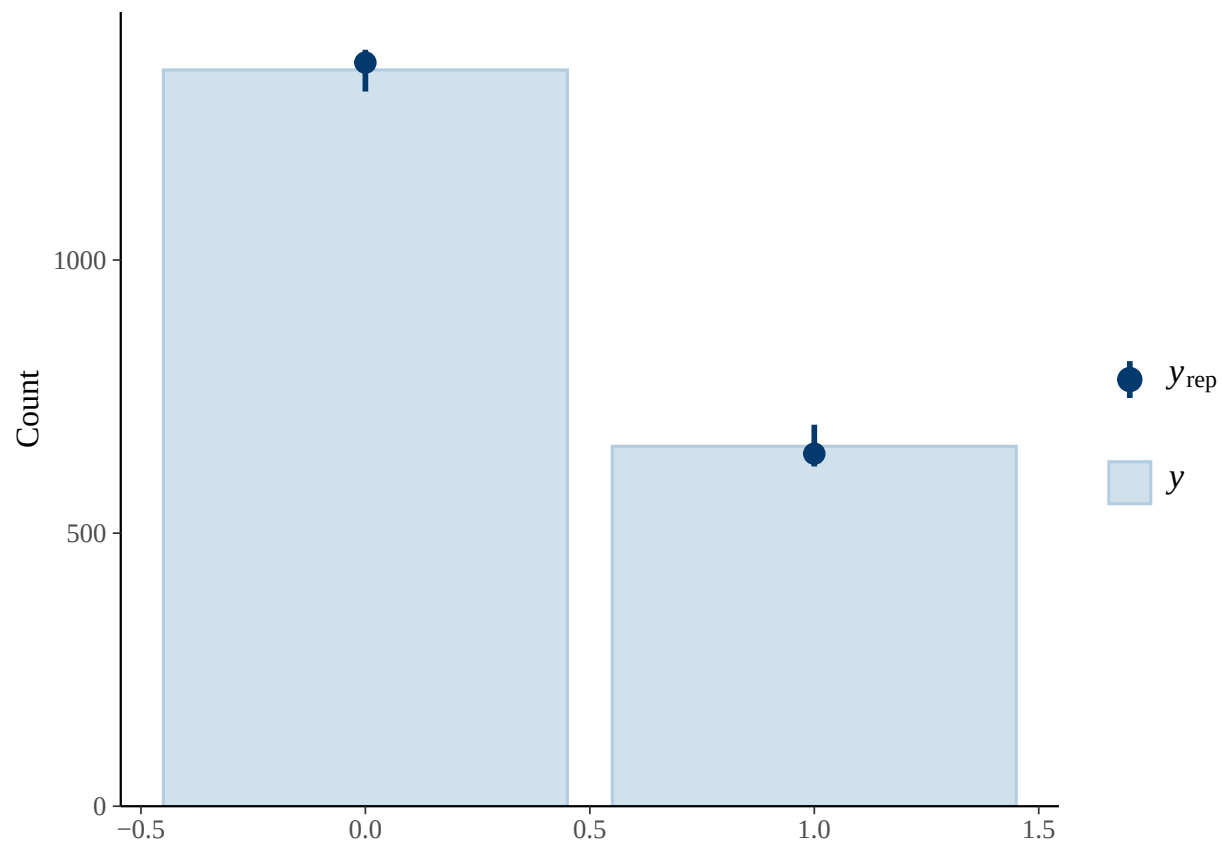


Chain

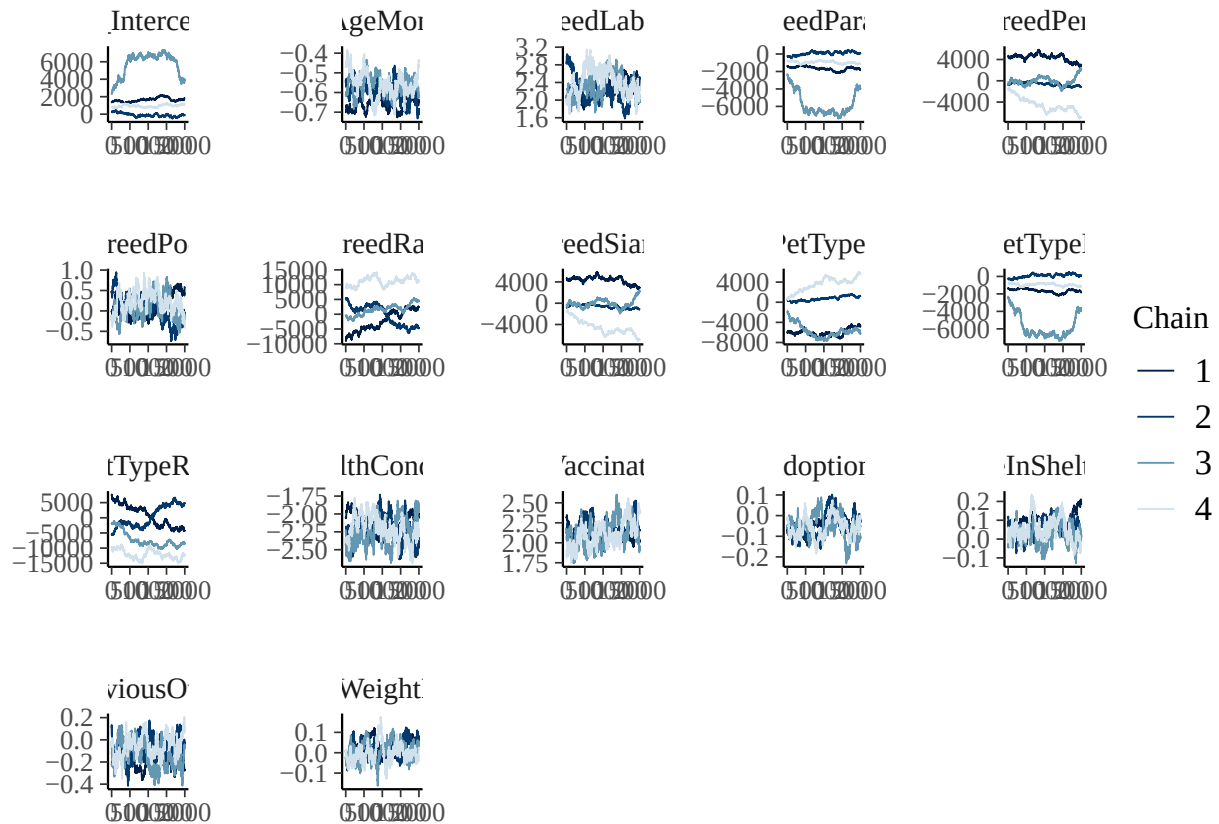
- 1
- 2
- 3
- 4



```
pp_check(model, type = "bars")
```

```
mcmc_plot(model, type = "trace")
```



```
bayes_R2(model)
```

```
##      Estimate   Est.Error    Q2.5    Q97.5
## R2 0.2851471 0.009972544 0.264232 0.303303
```

Optional: Re-train longer (once the model runs)

```
n_iter <- 4000
n_warmup <- 2000

model_long <- update(model, iter = n_iter, warmup = n_warmup, refresh = 100)
saveRDS(model_long, "bayes_model_fit_long.rds")
summary(model_long)
```

Load Saved Model

```
model <- readRDS("bayes_model_fit.rds")
summary(model)
```